

The Dynamics of Intra-Bank Networks and the Geographic Allocation of Credit

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Motivation & Research Questions

- Even in a fully integrated national financial system like the US, access to credit varies substantially across locations.
 - Two counties with identical demand for credit can face very different credit supply.
 - The reason is that the local cost of supplying credit depends on the availability of deposits and the presence of banks—both of which are determined by banks' geographic networks.
- This motivates two **Research Questions**.
 1. How much geographic allocation of credit reflects banks' networks?
 2. What determines those networks? In particular:
 - a. Differences between branch entry and loan-only entry?
 - b. Are there economies of scale or density?
 - c. How much network change through mergers rather than de novo?
 - d. How valuable is access to deposits as a source of network liquidity?

This Paper

- We study these questions by estimating a **model of bank competition** with two main features:
 - a. Banks compete for loans and deposits in local markets (counties), but these markets are interconnected by **banks' internal capital markets**: a bank's deposits in a county can be used to fund loans in other counties where the bank operates.
 - b. **Endogenous bank networks**. Banks choose the network of counties where they operate, with or without branches.
- Endogenizing bank networks is the main contribution of this paper.
- It matters for policy analysis.
 - Policies (mergers, branching deregulation, deposit insurance design, FinTech competition, monetary policy) reshape banks' incentives to reconfigure their networks.
 - **Treating networks as fixed** in the analysis of these policies can lead to serious misvaluations.

Related Literature

- **Credit access & internal capital markets**

- Petersen & Rajan (2002), Gilje (2019), Klapper et al. (2025); Coase (1937), Kashyap & Stein (1995)

- **Local deposit–loan synergies & market power**

- Kashyap, Rajan & Stein (2002), Mester, Nakamura & Renault (2007), Egan, Lewellen & Sunderam (2022); Drechsler, Savov & Schnabl (2017)

- **Structural models of bank competition**

- loan-side: Corbae & D’Erasmus (2019, 2021); deposit-side: Dick (2008), Egan, Matvos & Hortacsu (2016); branch geography: Aguirregabiria, Clark & Wang (2016), Oberfield et al. (2024)

- **Gap:** A key missing piece is an empirical framework that jointly models how internal capital markets connect deposits and lending across geography, and how banks endogenously build and reshape their branch and lending networks over time.

Roadmap

1. Data & Descriptive Evidence
2. Model
3. Estimation of Network Costs
4. Preliminary Results
5. Conclusions

1. Data & Descriptive Evidence

Data Sources

- **Bank-county-year panel:** 1998-2024; 3,146 counties; >9,000 banks.
 - Today's results: Period 1998-2010.
- **Main variables** at bank–county–year level: deposits, mortgage loans, branches, securitization rate.
- **Data sources:**

• Branches & deposits	FDIC Summary of Deposits
• Loans and securitization rate	HMDA
• Bank-level variables	Fed Call Reports
• Bank merger events	National Information Center
• County socioeconomics	Census Bureau and BEA

Expansion, Shutdowns, & Merger Attribution

Merger-implied: Expansion attributable to a merger.

Expansion type	All expansions		Merger-implied expansion	
	Events	Share	Events	Pr(Merger Type)
No presence → loan only	606,814	95.9%	9,759	1.6%
No presence → deposit/branch	14,274	2.3%	2,749	19.3%
Loan only → deposit/branch	11,901	1.9%	3,318	27.9%
All expansion events	632,989	100.0%	15,826	2.5%

Shutdown type	All shutdowns		Merger-implied shutdown	
	Events	Share	Events	Pr(Merger Type)
Loan only → no presence	586,207	96.6%	113,369	19.3%
Deposit/branch → no presence	18,656	3.1%	9,675	51.9%
Deposit/branch → loan only	1,861	0.3%	362	19.5%
All shutdown events	606,724	100.0%	123,406	20.3%

Expansion, Shutdowns, & County Adjacency

Adjacency: county has a neighbor in bank's pre-transition branch network.

Expansion type	Adjacent Events	Pr(Adjacency Type)
No presence → loan only	58,291	9.6%
No presence → deposit/branch	2,713	19.0%
Loan only → deposit/branch	5,487	46.1%
All expansion events	66,491	10.5%

Shutdown type	Adjacent events	Pr(Adjacency Type)
Loan only → no presence	528,566	90.2%
Deposit/branch → no presence	11,887	63.7%
Deposit/branch → loan only	1,078	57.9%
All shutdown events	541,531	89.3%

2. Model

Static Component of the Model

- **Static model of bank competition** for deposits and loans: *Aguirregabiria, Clark, & Wang (AER, 2025)*.
 - a. **County-level** price competition for differentiated deposits & loans
 - b. Spillovers between deposits and loans at the bank-county level (economies of scope).
 - c. Branches increase local demand for both loans and deposits.
 - d. A bank's securitization rate in a county reduces the bank's cost of lending there.
 - e. **Internal capital markets:** a bank's **total national deposits** reduce its lending cost in *every* county where it operates.
- In equilibrium, all counties are interconnected through the effect of total deposits on lending costs
- In this static model, **banks' networks** – the counties where a bank offers deposits and/or loans – are **taken as given**.

Dynamic Model: Endogenous Bank Networks

- Each bank j chooses two geographic networks across US counties:
 - Branch network:** $\mathbf{b}_{jt} = (b_{jmt} : m \in \mathcal{M})$, where $b_{jmt} = 1$ if bank j operates a branch in county m .
 - Loan network:** $\ell_{jt} = (\ell_{jmt} : m \in \mathcal{M})$, where $\ell_{jmt} = 1$ if bank j supplies loans in county m .
 - Combined network:** $\mathbf{x}_{jt} = (\mathbf{b}_{jt}, \ell_{jt})$
- The industry network configuration is

$$\mathbf{x}_t = \{(\mathbf{b}_{jt}, \ell_{jt}) : j \in \mathcal{J}\}$$

- Banks' deposit and lending networks are **endogenous equilibrium outcomes**.

Timing of Network and Market Decisions

- At the beginning of year t , banks observe:
 - the current industry network, $\mathbf{x}_t$
 - exogenous county and national states, $\mathbf{z}_t$
- Given current networks, banks compete in deposit & loan markets.
 - Static equilibrium determines prices, quantities, and variable profits:

$$\pi_j(\mathbf{x}_t, \mathbf{z}_t).$$

- Banks then choose their networks for the next year:

$$\mathbf{a}_{jt} = (a_{jmt}^b, a_{jmt}^\ell : m \in \mathcal{M}) = (\mathbf{b}_{j,t+1}, \boldsymbol{\ell}_{j,t+1})$$

- A bank can expand its network through:
 - De novo expansion:** choose individual counties in which to establish a branch or a loan-only presence.
 - Acquisition:** acquire another bank and inherit its entire networks.

Network Costs and Economies of Density

- a. **Fixed costs** of operating a branch or loan-only presence in county

$$FC_{jmt}^b = \gamma_{0j}^b + \gamma_1^b \text{Density}_{jmt} + \gamma_2^b z_{mt}$$

$$FC_{jmt}^\ell = \gamma_{0j}^\ell + \gamma_1^\ell \text{Density}_{jmt} + \gamma_2^\ell z_{mt}$$

- b. **De novo Entry costs** of adding a new branch or entering a county through loan-only channels:

$$EC_{jmt}^{0 \rightarrow b} = \eta_{0j}^{0 \rightarrow b} + \eta_1^{0 \rightarrow b} \text{Density}_{jmt} + \eta_2^{0 \rightarrow b} z_{mt}$$

$$EC_{jmt}^{0 \rightarrow \ell} = \eta_{0j}^{0 \rightarrow \ell} + \eta_1^{0 \rightarrow \ell} \text{Density}_{jmt} + \eta_2^{0 \rightarrow \ell} z_{mt}$$

$$EC_{jmt}^{\ell \rightarrow b} = \eta_{0j}^{\ell \rightarrow b} + \eta_1^{\ell \rightarrow b} \text{Density}_{jmt} + \eta_2^{\ell \rightarrow b} z_{mt}$$

- c. **Acquisition / Merging costs** If bank j acquires bank k :

$$EC_{jt}^{acq \rightarrow k} = \eta_{0j}^{acq} + \eta_1^{acq} N_{jkt}^{new} + \eta_2^{acq} Size_{kt} + \eta_3^{acq} Overlap_{jkt} + \eta_4^{acq} z_t$$

Dynamic Game of Network Competition

- Banks choose networks to maximize the present value of profits:

$$\Pi_j(\mathbf{a}_{jt}, \mathbf{x}_t, \mathbf{z}_t) = \pi_j(\mathbf{x}_t, \mathbf{z}_t) - FC_j(\mathbf{x}_{jt}, \mathbf{z}_t) - EC_j(\mathbf{a}_{jt}, \mathbf{x}_{jt}, \mathbf{z}_t)$$

- Banks' networks result from a **Markov Perfect Equilibrium**.
 - Banks' strategies are functions of payoff-relevant state variables:

$$\mathbf{a}_{jt} = \sigma_j(\mathbf{x}_t, \mathbf{z}_t)$$

- The optimal network decision satisfies the Bellman equation:

$$V_j(\mathbf{x}_t, \mathbf{z}_t) = \max_{\mathbf{a}_{jt} \in \mathcal{A}} \{ \Pi_j(\mathbf{a}_{jt}, \mathbf{x}_t, \mathbf{z}_t) + \beta \mathbb{E} [V_j(\mathbf{x}_{t+1}, \mathbf{z}_{t+1}) \mid \mathbf{a}_{jt}, \mathbf{x}_t, \mathbf{z}_t] \}.$$

The expectation incorporates the equilibrium network choices of competing banks.

3. Estimation of Network Costs

Revealed Preference

- The (static) **equilibrium variable-profit function**, $\pi_j(\mathbf{x}_t, \mathbf{z}_t)$, can be computed for any counterfactual network configuration $\mathbf{x}_t^*$.
 - Structural demand, cost parameters are estimated in ACW (2025).
 - Computing an equilibrium requires repeatedly solving Bertrand equilibria across thousands of counties. We develop an efficient algorithm that completes these calculations in only a few seconds.
- Fixed and entry costs are unknown but parametric: $NC_{jt}(\cdot, \boldsymbol{\theta})$.
- By Revealed Preference, for any $\mathbf{x}_{j,t+1}^* \neq \mathbf{x}_{j,t+1}$:

$$\begin{aligned} & -NC_{jt}(\mathbf{x}_{jt} \rightarrow \mathbf{x}_{j,t+1}, \boldsymbol{\theta}) + \beta \mathbb{E}_t[V_j(\mathbf{x}_{t+1}, \mathbf{z}_{t+1})] \\ \geq & -NC_{jt}(\mathbf{x}_{jt} \rightarrow \mathbf{x}_{j,t+1}^*, \boldsymbol{\theta}) + \beta \mathbb{E}_t[V_j(\mathbf{x}_{j,t+1}^*, \mathbf{x}_{-j,t+1}, \mathbf{z}_{t+1})] \end{aligned}$$

- Every perturbation $\mathbf{x}_{j,t+1}^*$ (potentially millions of them) generates one inequality in observables and cost parameters.

Finite Dependence Property

- To get around solving for the value function V_j , we exploit the **one-period finite dependence property** of this dynamic game.
- We compare factual choice sequences to one-period counterfactual perturbations:

- **factual:** $\mathbf{x}_{jt} \rightarrow \mathbf{x}_{j,t+1} \rightarrow \mathbf{x}_{j,t+2}$
- **counterfactual:** $\mathbf{x}_{jt} \rightarrow \mathbf{x}_{j,t+1}^* \rightarrow \mathbf{x}_{j,t+2}$

- Revealed Preference implies the **Euler-like inequality**:

$$\begin{aligned} & -NC_j(\mathbf{x}_{jt} \rightarrow \mathbf{x}_{j,t+1}, \boldsymbol{\theta}) + \beta \mathbb{E}_t [\pi_j(\mathbf{x}_{t+1}) - NC_j(\mathbf{x}_{j,t+1} \rightarrow \mathbf{x}_{j,t+2}, \boldsymbol{\theta})] \\ \geq & -NC_j(\mathbf{x}_{jt} \rightarrow \mathbf{x}_{j,t+1}^*, \boldsymbol{\theta}) + \beta \mathbb{E}_t [\pi_j(\mathbf{x}_{j,t+1}^*, \mathbf{x}_{-j,t+1}) - NC_j(\mathbf{x}_{j,t+1}^* \rightarrow \mathbf{x}_{j,t+2}, \boldsymbol{\theta})] \end{aligned}$$

- That is, the changes in variable profits $\pi_j(\mathbf{x}_{t+1}) - \pi_j(\mathbf{x}_{j,t+1}^*, \mathbf{x}_{-j,t+1})$ provide informative bounds for the Network Cost parameters.

Perturbations: $\Delta_{j,t+1} = \mathbf{x}_{j,t+1}^* - \mathbf{x}_{j,t+1}$

- We consider **local, single-county, single-bank** perturbations.
- We consider **Contraction** and **Expansion** perturbations.
- For given bank j , county m , year t .

Type	Perturbation	
Contraction	$b \rightarrow \ell$	Shut down branch, keep local loan link
Contraction	$\ell \rightarrow 0$	Shut down a loan-only location
Contraction	$b \rightarrow 0$	Shut down local deposits <i>and</i> loans
Expansion	$\ell \rightarrow b$	Add a branch where the bank already lends
Expansion	$0 \rightarrow \ell$	Add a loan-only location, no branch
Expansion	$0 \rightarrow b$	Add a branch <i>and</i> enter loan market, new county

Intrinsic Interest of these Perturbations

- The **effects of these perturbations** on the static equilibrium, such as changes in variable profits $\pi_j(\mathbf{x}_{t+1}) - \pi_j(\mathbf{x}_{j,t+1}^*, \mathbf{x}_{-j,t+1})$, have economic interest by themselves.
- We use data from these perturbations to study the following empirical questions:
 - **Determinants of network value.** What predicts the profit value of a bank adding or dropping a county?
 - **Geographic reallocation of credit.** Counterfactual branch openings/closings let us measure how lending in other counties responds—an elasticity of interregional liquidity reallocation.
 - **Internal capital markets.** Bank-level measure of how much lending elsewhere shifts per dollar of deposits lost locally through simulated deposit-market exits.

4. Preliminary Results

Profit Changes from Local Perturbations

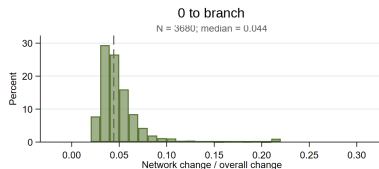
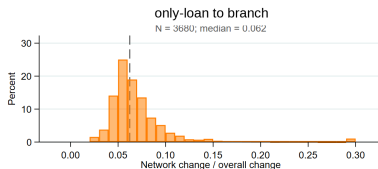
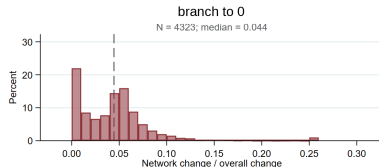
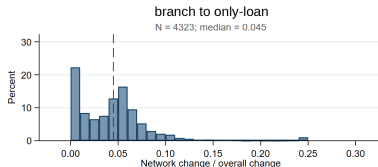
Mean [Median] Variable-Profit change, in \$000.

CF	Number bank-years	Overall profit change	At Target-county profit change	Rest Network profit change
Contractions:				
$b \rightarrow \ell$	4,323	-105.2 [-52.1]	-99.4 [-49.7]	-5.8 [-1.5]
$b \rightarrow 0$	4,323	-105.4 [-52.1]	-99.6 [-49.8]	-5.8 [-1.5]
$\ell \rightarrow 0$	4,125	-0.5 [-0.2]	-0.5 [-0.2]	0.0 [0.0]
Expansions:				
$\ell \rightarrow b$	3,680	37.5 [24.3]	34.8 [22.4]	2.7 [1.6]
$0 \rightarrow \ell$	3,680	0.5 [0.2]	0.5 [0.2]	0.0 [0.0]
$0 \rightarrow b$	3,680	24.5 [16.5]	23.5 [15.7]	1.1 [0.8]

Distribution of Network Spillovers Ratio

Heterogeneity in Network Spillover Shares

Ratios winsorized within each counterfactual at the 1st and 99th percentiles



Dashed lines show medians. Branch to only-loan and branch to 0 are contractions; only-loan to branch and 0 to branch are expansions.

What Predicts Expansion Network Value?

Dependent variable: log total profit change. Year-2000

	$\ell \rightarrow b$	$0 \rightarrow \ell$	$0 \rightarrow b$
Log ($M_m s_{0m}$) in target-market	0.534*** (0.011)	0.642*** (0.011)	0.488*** (0.012)
Normalized incumbent deposit HHI	0.458*** (0.092)	-0.112 (0.065)	0.370*** (0.070)
Log distance to network center	-0.110*** (0.023)	-0.056** (0.022)	-0.154*** (0.024)
Network density	-0.110*** (0.027)	0.013 (0.024)	-0.143*** (0.026)
Log bank's total deposits	0.055*** (0.014)	0.251*** (0.012)	0.057*** (0.013)
Log number branch counties	0.011 (0.023)	0.010 (0.021)	0.169*** (0.023)
Log number loan-only counties	0.089*** (0.013)	0.065*** (0.012)	-0.031** (0.013)
Headquarters-state fixed effects	Yes	Yes	Yes
Observations	3,675	3,659	3,680
R^2	0.529	0.720	0.513

Estimation of Network Entry Costs

- $\log(\text{Entry Cost})$. Parameters are elasticities.
- Year 2000. No shadow banks. Shutdown perturbations only.

	Adjacent	100 km density
Form of entry: $\ell \rightarrow b$ ($0 \rightarrow \ell$ omitted)	5.926*** (0.033)	5.967*** (0.032)
Form of entry: $0 \rightarrow b$ ($0 \rightarrow \ell$ omitted)	5.917*** (0.032)	5.970*** (0.031)
Scale Economies: Log(branch counties)	-0.719*** (0.038)	-0.766*** (0.043)
Scale Economies: Log(loan-only counties)	-0.092*** (0.029)	-0.036 (0.033)
Density Econ.: Log(Adjacent branch counties)	-0.176*** (0.041)	
Density Econ: Log(Adjacent loan-only counties)	0.324*** (0.036)	
Density Econ.: Log(100km branch density)		-0.198*** (0.057)
Density Econ.: Log(100km loan-only density)		0.163*** (0.043)
Bank Fixed Effects	Yes	Yes
Observations	12,760	12,760

Preliminary Conclusions

1. **Branches vs. loan-only.** Branches have substantially greater variable-profit than loan-only links. However, loan-only entry & exit are much cheaper, providing greater flexibility.
2. Substantial **economies of scale and density in entry cost** for branching but much smaller or none for loan-only.
3. Although mergers account for only 2.5% of expansions, they account for more than 20% of contractions: **substantial geographic reallocation following bank mergers.**
4. **Next steps** are to estimate and validate network costs, incorporate mergers, and study geographic credit access and policy counterfactuals.

Thank you!